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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Load the image from Colab environment
image_path = "image1.jpg"
image = cv2.imread(image_path)

if image is not None:
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # 1. Calculate gradients using Sobel operators
    grad_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
    grad_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

    # 2. Calculate the magnitude of the gradient (Gradient Mask)
    abs_grad_x = cv2.convertScaleAbs(grad_x)
    abs_grad_y = cv2.convertScaleAbs(grad_y)
    gradient_mask = cv2.addWeighted(abs_grad_x, 0.5, abs_grad_y, 0.5, 0)

    # 3. Sharpening: Original + Gradient Mask
    # We add the edges back to the original image
    sharpened = cv2.add(gray, gradient_mask)

    # Display results
    print("Original Grayscale:")
    cv2_imshow(gray)
    print("\nGradient Mask (Edges):")
    cv2_imshow(gradient_mask)
    print("\nSharpened Image (Gradient Masking):")
    cv2_imshow(sharpened)

    # Save the result
    cv2.imwrite('gradient_sharpened.jpg', sharpened)
else:
    print(f"Error: '{image_path}' not found. Please ensure the image is up.
```

Original Grayscale:



Gradient Mask (Edges):



Sharpened Image (Gradient Masking):

